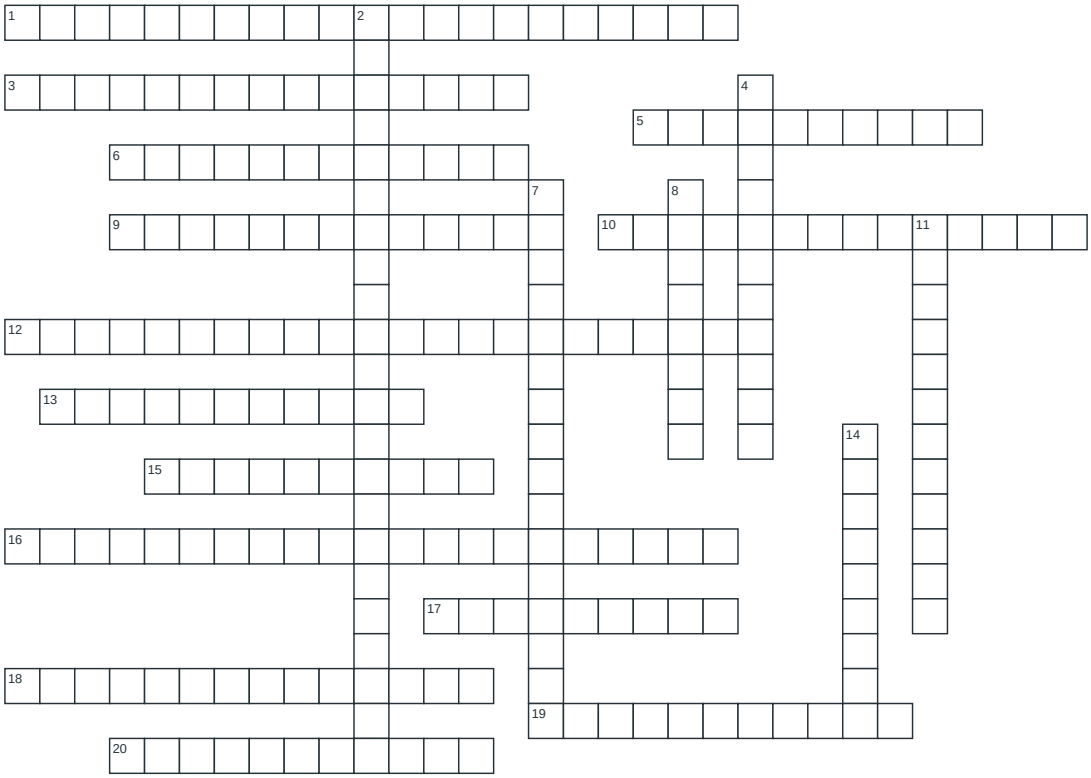
Stars and Stellar Life Cycles Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

1. An explosion associated with collapse of a massive star's core

3. Expelled gas surrounding the remnant of a lower-mass star.

5. A dense stellar remnant that gradually cools

6. Rapid inward contraction of a massive star’s core.

9. Very large, cool late stage of a massive star.

10. Compact object left after a star sheds or collapses its outer layers.

12. Balance between inward gravity and outward pressure in a stable star.

13. Group of stars held near one another by gravity.

15. Total energy emitted by a star per unit time

16. Contraction of matter under its own gravitational attraction.

17. Early stage in star formation before sustained hydrogen fusion begins

18. Nuclear process that combines hydrogen nuclei to form helium and release energy.

ACROSS (continued)

19. Material in a star that can release energy through nuclear reactions.

20. Extremely dense remnant that can form after a massive stellar core collapses.

DOWN

2. Formation of atomic nuclei within stars through nuclear reactions.

4. Amount of matter in a star, a major factor in its evolution.

7. Changes in a star’s structure and energy production over its lifetime.

8. Expanded late stage of a lower- or intermediate-mass star.

11. Long-lived stellar stage when a star fuses hydrogen in its core.

14. Region where gravity is so strong that light cannot escape from within its event horizon.

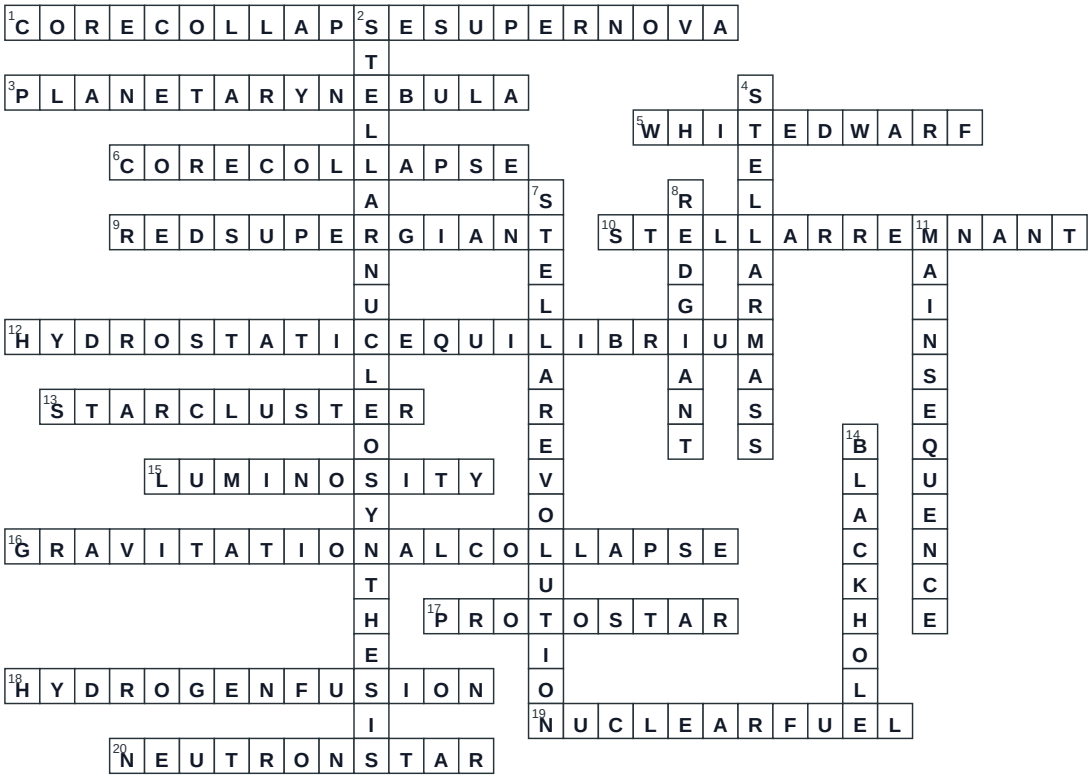
© 2026 Teach Arcade

Stars and Stellar Life Cycles Crossword

Hard | 20 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

1. Core-collapse Supernova
3. Planetary nebula
5. White Dwarf
6. Core collapse
9. Red supergiant
10. Stellar remnant
12. Hydrostatic equilibrium
13. Star cluster
15. Luminosity
16. Gravitational collapse
17. Protostar

ACROSS (continued)

18. Hydrogen fusion
19. Nuclear fuel
20. Neutron star

DOWN

2. Stellar nucleosynthesis
4. Stellar mass
7. Stellar evolution
8. Red giant
11. Main sequence
14. Black hole

© 2026 Teach Arcade